G0 rung 0 — VERDICT: FAIL (failed: iv_grows_alone; SCALE untested)

run `20260919-164355` · genome `runs/g0b-20260919-164355/G0.genome` · PREREG docs/G0-PREREG.md §1–§11 · every number below is read from results.json

---

The nine marks, bars as written before the run (§5), predictions of §7 beside each

mark	bar	read	pass	§7 predicted
(i) it answers	Q1,Q2 ≥ .80 · Q3 ≥ .85 · Q4 ≥ .90	Q1 0.995 · Q2 1.00 · Q3 0.995 (ceiling 0.998) · Q4 0.995	True	likely PASS
(ii) the meeting is real	T-shuffle, T-nowire Q1 ≤ .20	T-shuffle Q1 band [0.0, 0.132, 0.055] · T-nowire Q1 0.000 (abstain {'novel': 1, 'noedge': 219, 'ratio': 0})	True	PASS
(iii) growth matters	T-frozen ≥ 30 pp under (i)	T-frozen Q1 0.191 vs design 0.995	True	PASS
(iv) it grows on its own	≥ 50 born · 0 refused · 5-thing < full world · ≥ 5 % pruned · 0 by disuse	born 86 · refused 0 · 5-thing 21 vs full world 43 at the end of development · pruned 1/72 = 1.4 % (cross-thing born 2, cross-thing pruned 1) · disuse 0	False	births 100–300; pruning may FAIL the 5 % bar
(v) it keeps what it learned	within 5 pp after arrival 2	Q1 first-10 at 600 0.995 → at 900 0.995 (+0.0 pp)	True	PASS
(vi) it knows what it does not know	plant majority answered · null majority abstains · Q5 ≥ .80 · false ≤ .20	plant zog answered 0.55 (as {'zog': 11}) · null vex abstains 0.90 · Q5 spiral abstains 1.00 (by {'novel': 20, 'noedge': 0, 'ratio': 0}; answered as {}) · false abstention on seen 0.005 (by {'novel': 1, 'noedge': 0, 'ratio': 0})	True	the uncertain one
(vii) replay	two lives byte-identical	True (sha256 cfbc80b3231f8755…)	True	PASS by construction
(viii) cost	≤ 300 s, one process	2.33 s	True	—
(ix) not luck	top quartile of 20 random genomes	design composite 0.997 ranks 1 of 21 (best random 0.959)	True	unknown
SCALE	—	0 events	UNTESTED	—

The design life, every number

- Gate G-a wrote THETA_N = 0.968, THETA_M = 0.9986; retina `thresh0.25 centre tower:eye-tower-g2 sqrt unit` (112 numbers); §2's strict rule NOT MET, §11's amended rule met (see the genome file's last line).
- Toy (circle, bar; 2 × 150 moments): Q1 1.000 → PASS.
- The eye: the numpy forward vs the sealed E09d tower's own 2356-frame gate features — max |diff| 0.00e+00, 2356 of 2356 within 1e-4, planted shifted frame caught True; the gene's tower eye-tower-g2 (sha 6abc6d7553d8b8c7…, 112 features) runs the same arithmetic on its seal-checked bytes.
- Things: circle, square, triangle, cross, ring, bar, dot, star, arrow, wave, diamond at 32×32; plant showings 5 (gate: within-thing share ≥ THETA_N 0.503); closest pair circle/diamond between-p95 0.963.
- At the end of development (moment 660): cells 43 sight + 11 words; Q1 0.995 Q2 1.00 Q3 0.995 Q4 0.995; abstain by {'novel': 1, 'noedge': 0, 'ratio': 0}.
- Edges by the ledger's reading: cross-thing born 2 of 75, cross-thing pruned 1 of 1 pruned in all (G0b §3 diagnostic; a same-thing edge is one whose two cells were born on the same thing).

- At the end of arrival 2 (moment 960): cells 70 + 16 (born novelty 70/16, error 0/0 — the error road FIRED 27 times and every birth was refused as redundant (§11 am. 4); refused by budget 0, by arrival [0, 0, 0]); edges alive 71, born 72, pruned 1; merged cells 0/0; removed for disuse 0.
- Q1 per thing at 900: {"circle": 1.0, "square": 1.0, "triangle": 1.0, "cross": 1.0, "ring": 0.95, "bar": 1.0, "dot": 1.0, "star": 1.0, "arrow": 1.0, "wave": 1.0, "diamond": 1.0}; arrival-2 things Q1 1.000 per thing {"moon": 1.0, "house": 1.0, "tree": 1.0, "cup": 1.0, "key": 1.0}.
- Q2 detail (word → the thing of the cell it lit): {"circle": "circle", "square": "square", "triangle": "triangle", "cross": "cross", "ring": "ring", "bar": "bar", "dot": "dot", "star": "star", "arrow": "arrow", "wave": "wave", "diamond": "diamond"}.
- Cells per thing (alive, single-thing): {"circle": 2, "square": 3, "triangle": 5, "cross": 7, "ring": 3, "bar": 5, "dot": 2, "star": 1, "arrow": 5, "wave": 5, "diamond": 5, "moon": 2, "house": 6, "tree": 5, "cup": 11, "key": 3}; multi-thing sight cells 0; answers from multi-word cells 0.
- After the plant (moment 965): Q1 on the first things 0.995; ledger-corruption plant: hash unchanged True.
- Time: design life 2.33 s; replay identical True.

Twins (§4) — each blind to one thing

- **T-shuffle-5**: Q1 0.000 (abstain {'novel': 0, 'noedge': 0, 'ratio': 95}) · Q2 0.27 · Q3 0.464 · Q4 0.168 · cells 93+16 (error births 31/0, redundant refused 69) · edges alive 442 born 506 pruned 56 (11.1 %) · merged 7/0 · refused budget 0 frozen 0 · 1.93 s
- **T-shuffle-6**: Q1 0.132 (abstain {'novel': 0, 'noedge': 0, 'ratio': 69}) · Q2 0.09 · Q3 0.541 · Q4 0.523 · cells 90+16 (error births 26/0, redundant refused 62) · edges alive 408 born 464 pruned 51 (11.0 %) · merged 5/0 · refused budget 0 frozen 0 · 1.92 s
- **T-shuffle-7**: Q1 0.055 (abstain {'novel': 0, 'noedge': 0, 'ratio': 102}) · Q2 0.09 · Q3 0.514 · Q4 0.400 · cells 90+16 (error births 27/0, redundant refused 74) · edges alive 407 born 467 pruned 53 (11.3 %) · merged 7/0 · refused budget 0 frozen 0 · 1.91 s
- **T-frozen**: Q1 0.191 (abstain {'novel': 173, 'noedge': 0, 'ratio': 0}) · Q2 0.45 · Q3 0.591 · Q4 0.191 · cells 5+5 (error births 0/0, redundant refused 0) · edges alive 5 born 5 pruned 0 (0.0 %) · merged 0/0 · refused budget 0 frozen 1475 · 1.35 s
- **T-nowire**: Q1 0.000 (abstain {'novel': 1, 'noedge': 219, 'ratio': 0}) · Q2 0.00 · Q3 0.500 · Q4 0.000 · cells 70+16 (error births 0/0, redundant refused 0) · edges alive 0 born 0 pruned 0 (0.0 %) · merged 0/0 · refused budget 0 frozen 0 · 1.68 s
- **five-thing life** (600 moments, 120/thing): cells born 21 vs the full world's 43 at the end of development; Q1 1.000.

Twenty random genomes (§4, E29's control) — composite = mean(Q1..Q4) at 900

seed	composite	Q1	Q2	Q3	Q4	theta_n	misfires	budget	kwta	lr	prune	theta_m	stage1	abstain	cells	refused
19	0.959	0.895	1.00	0.950	0.991	0.954	4	334	4	0.023	0.063	0.971	236	0.76	44	0
7	0.895	0.800	0.91	0.918	0.955	0.940	6	484	10	0.071	0.061	0.986	378	0.90	36	0
12	0.874	0.809	0.91	0.900	0.877	0.964	3	98	3	0.058	0.071	0.923	245	0.56	40	0
1	0.833	0.759	0.91	0.877	0.786	0.966	5	76	2	0.285	0.063	0.942	189	0.70	37	25
8	0.816	0.709	1.00	0.845	0.709	0.984	3	182	4	0.238	0.174	0.939	288	0.68	85	0
13	0.709	0.395	0.82	0.686	0.936	0.919	7	102	10	0.082	0.016	0.994	359	0.50	26	0
11	0.649	0.373	0.55	0.691	0.986	0.745	2	493	8	0.013	0.030	0.992	53	0.56	154	0
5	0.622	0.345	0.64	0.677	0.827	0.896	7	402	7	0.089	0.012	0.938	268	0.52	21	0
2	0.351	0.000	0.27	0.500	0.632	0.646	3	360	10	0.032	0.120	0.972	335	0.53	56	0
15	0.273	0.164	0.18	0.582	0.164	0.900	6	231	5	0.018	0.115	0.914	373	0.67	22	0
4	0.182	0.000	0.09	0.500	0.136	0.751	8	756	12	0.029	0.122	0.937	291	0.59	24	0
20	0.168	0.000	0.09	0.500	0.082	0.726	3	722	2	0.159	0.082	0.907	357	0.98	9	0
10	0.147	0.082	0.00	0.495	0.009	0.602	8	645	10	0.049	0.103	0.913	311	0.91	5	0
3	0.143	0.000	0.00	0.500	0.073	0.616	1	186	10	0.177	0.020	0.943	325	0.58	28	1184

6	0.125	0.000	0.00	0.500	0.000	0.668	5	760	5	0.115	0.198	0.963	178	0.66	11	0
9	0.125	0.000	0.00	0.500	0.000	0.641	7	135	8	0.234	0.143	0.991	169	0.95	3	0
14	0.125	0.000	0.00	0.500	0.000	0.677	7	116	9	0.259	0.129	0.954	60	0.85	4	0
16	0.125	0.000	0.00	0.500	0.000	0.711	5	625	2	0.108	0.125	0.902	215	0.92	7	0
17	0.125	0.000	0.00	0.500	0.000	0.579	7	393	7	0.114	0.044	0.938	297	0.80	1	0
18	0.125	0.000	0.00	0.500	0.000	0.852	4	702	4	0.029	0.194	0.956	358	0.78	13	0

The design genome's composite 0.997 ranks 1 of 21; the best random genome is seed 19 at 0.959.

What the design life did and did not exercise, read from the numbers

- Error births 0/0 (the road fired 27 times; every birth refused as redundant, §11 am. 4); merges 0/0 (THETA_M 0.9986); edges pruned 1 of 72; cross-thing edges born 2 of 75, cross-thing pruned 1; scale 0. THETA_N 0.968 against between-thing p95 0.903, max 0.970: 2 cross-thing edges were born and 1 of them pruned — the SHRINK genes had that much work and did it.
- In T-shuffle-5 (random pairings) 506 edges were born and 56 pruned (11.1 %), with 31 error births and 7 merges: the PRUNE, error-birth and MERGE genes work when there is something to prune, birth or merge; the design world gives them little.
- The plant: «zog» shown 5 times birthed 3 cells; 0.55 of its 20 held-out renderings were answered «zog», 0.45 abstained (by {'novel': 9, 'noedge': 0, 'ratio': 0}); the gate's within-thing share at THETA_N was 0.503 — the showings formula assumed independent cells, and same-thing cells are not independent.
- Q5: the spiral abstains at 1.00 (by {'novel': 20, 'noedge': 0, 'ratio': 0}; answered as nothing) against its 0.80 bar; the spiral was never in gate G-a's measurement, so the threshold carries no guarantee for a never-shown thing.

What this verdict does and does not say

- The truth never entered the organism: `genome/machinery.py` imports neither `world` nor `questions` and carries no label field; the ledger is the runner's, built from the stream (clause 12; the hash check proves only that the runner did not touch the organism).
- SCALE is untested here (0 events in 965 moments). §2's strict gate rule was not met; the amended rule was, at the price of several cells per thing — the cells-per-thing line above is that price, measured.
- No sentences, no reasoning, no reward, no sleep, no evolution (§6). Rung 1 is his pictures through the folder door; rung 2 a lineage, only after rung 1.